

Institute of Actuaries of India

Subject CT7 – Business Economics

May 2012 Examinations

INDICATIVE SOLUTIONS

Solutions 1 to 30 :

1. A
2. B
3. D
4. B
5. A
6. D
7. D
8. D
9. A
10. A
11. A
12. B
13. D
14. D
15. D
16. B
17. B
18. A
19. A
20. C
21. D
22. A
23. A
24. C
25. B
26. B
27. C
28. C
29. C
30. C

[Q.No. 1 to 30 =45 Marks]

Solution 31 :

The Phillips curve shows an inverse relationship between unemployment and inflation. Phillips curves are drawn on a graph with inflation on the vertical axis and unemployment on the horizontal axis.

Accelerationist view – No long run-trade off between inflation and unemployment and LRPC is vertical at the natural rate of unemployment because expectations are augmented every time inflation rises and nominal wages adjust to keep real wages constant. But there is a short-run trade off possible while the expectations of future inflation take time to adapt.

Rational expectations – Even the SRPC is vertical so there is no short-term trade off either. Individuals and firms use all the economic information available to them to make their decisions. This theory also assumes that prices and wages are flexible so that markets clear quickly in response to changes in expectations. Economy always operates at the natural rate of unemployment (even in the short run).

[5]

Solution 32 :

The three types of withdrawals are:

1. *Net savings (S)*. Savings is that part of income that is not spent.

Net savings = savings – borrowing – drawing on past savings.

2. *Net taxes (T)*. Taxes on factor incomes, *eg* income tax, and taxes on consumer expenditure, *eg* VAT; net of transfer payments like subsidies.

Net taxes = taxes – transfer payments.

3. *Imports (M)*. Expenditure on goods and services produced abroad leaks out of the domestic economy.

The three types of injections are:

1. *Investment (I)*. This is the expenditure on capital goods such as plant and equipment. It also includes the building up of stock.
2. *Government expenditure (G)*. This is expenditure on goods and services such as roads and schools. (This does *not* include spending on transfer payments.)
3. *Exports (X)*. This is expenditure by foreign residents on goods and services produced domestically.

[6]

Solution 33 :

(a)

If the Chinese government is pegging the exchange rate at an abnormally low value, then it must be expanding the domestic money supply more than it otherwise would to meet any increased demand for 'yuan'. Higher money tends to raise the price level.

However, China may be able to forestall this increase in prices by sterilizing its foreign currency intervention through open market operations or reduction in government spending. Thus, Chinese price level 'may' or 'may not' increase depending on whether the government intervenes in the domestic market using fiscal policy instruments.

(b)

If interest rates are higher in country A than in country B, with a fixed exchange rate money from abroad will flow in from country B to country A. To maintain the fixed exchange rate the government will have to sell the domestic currency to meet the demand for money. This increases the money supply and pushes interest rates down, towards the level in country B. Thus, an independent monetary policy is not possible under fixed exchange rates.

With expansionary fiscal policy, private sector consumption and investment is crowded out by higher interest rates and higher prices. Under fixed exchange rates with perfect capital mobility interest rates cannot be allowed to rise, which limits the amount of crowding out that can occur. Hence fiscal policy is more effective with fixed exchange rates.

[7]

Solution 34 :

a) Fiscal policy measures to increase output:

1) An increase in government spending: This increases output through the *multiplier effect*. If *mpc* in the economy is 0.75 then multiplier will be 4. Thus an increase in government spending by 1 million rupees increases national income by 4 million rupees. The increase in income will increase the demand for investment through *accelerator effect* and this further increases the national income.

2) A reduction in taxation: This increases the disposable income with the households and firms, thus increasing their spending and demand for goods and investment. If *mpc* is .75, a decrease in taxes by 1 million rupees increases spending by .75 million rupees and through the multiplier effect increases income by 3 million rupees.

The above fiscal measures will be ineffective when:

- 1) A pure fiscal expansion can result in crowding out whereby government spending diverts resources away from the private sector e.g through higher interest rates. If this crowding out is total the level of national income may not increase
- 2) Reduction in taxes- when *mpc* is low the multiplier effect will be low and hence increase in government spending and reduction in taxes will have a smaller effect on income.

b) Monetary policy measures to increase liquidity in the economy:

- 1) Increasing the money supply: The central bank can increase the money supply in the economy by buying government securities in the open market, by increasing the amount it is willing to lend to the banks, altering the level of liquid assets held by the bank and by decreasing the minimum reserve ratio.
- 2) Decreasing interest rates: The central bank can decrease the interest rate at which it lends to the banks (through the repo rate and the bill rediscount rate). This will decrease the interest rate at which the banks lend and hence will increase demand for money and hence liquidity.

The above monetary measures will be ineffective when:

- 1) The money supply may not translate into liquidity at the hands of public because credit creation is determined by the demand for borrowing. If people do not wish to borrow, banks will not be able to create credit.

2) The demand for money is insensitive to interest rates. In this case any attempt to increase the demand for money will require a large fall in interest rates.

c) Recent steps taken by RBI to control inflation:

- 1) Increase in the minimum reserve ratio
 - 2) Increase in repo and reverse repo rate
 - 3) Increase in Statutory liquidity requirement of the banks
- (Any two of the above can be mentioned).

[11]

Solution 35 :

A. Opportunity cost = explicit + implicit costs = \$25,000 + \$27,000 (\$20,000 + \$7,000) = \$52,000 per month.

B. Brown's costs of production (= \$52,000/month) exceed his revenues by \$17,000 (= 52,000 – 35,000). Rather than lose \$17,000 per month, Brown could rent his rig (and receive \$20,000 per month) and drive trucks for another firm (and earn \$7,000 per month). With this use of his resources he would earn \$27,000 per month.

[3]

Solution 36 :

Break-even quantity = $[2,00,000 / (600 - 500)] = 2,000$

The maximum quantity demanded at the price of 600 would be = $8,000 - 10(600) = 2,000$

Thus, at most Mr. Rodan could break-even. So he decided not to start the business.

[2]

Solution 37 :

A.

Dominant strategy is a strategy or action that always provides the best outcome no matter what decisions rivals make.

Alpha has no dominant strategy in above game because for Alpha there is no single strategy which always gives the best outcome irrespective of Beta's decision/action. Against 'high price' strategy of Beta, 'Low price' strategy gives the best to Alpha; against 'Moderate price' strategy of Beta, 'moderate price' strategy gives the best to Alpha; and against 'Low price strategy' of Beta, 'High price' strategy gives the best outcome to Alpha.

B.

Nash equilibrium is a set of actions for which all players are choosing their best actions given the actions chosen by their rivals. Or Nash equilibrium is the position resulting from everyone making their optimal decision based on their assumptions about their rival's decision.

There are two Nash equilibriums in this game, 'Moderate Price, Moderate price' and 'High Price, Low price' with payoff of '1250, 1100' and '800, 1500', respectively.

[5]

Solution 38 :

'Favourable odds' is where on average the gambler will gain. If, for example, a gambler is offered odds of 10 to 1 on the throw of a dice, then for \$1 bet he would get nothing if he lost, but he would get \$10 if his number came up. Since his number should come up on average one time in every six, on average he will gain. If the odds were 6 to 1, they would be fair odds. On average, the gambler would break even. If, odds were less than 6 to 1, they would be unfavourable odds. On average, the gambler would lose.

Three possible categories of attitude towards risk are risk neutral, risk loving, and risk averse. A risk neutral person will take a gamble if the odds are favourable; not take a gamble if the odds are unfavourable; and be indifferent about taking a gamble if the odds are fair. Risk loving person is prepared to take a gamble even if the odds are unfavourable. Risk-averse person may not be prepared to take a gamble even if the odds are favourable.

[6]

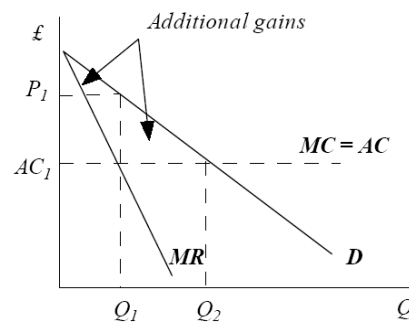
Solution 39 :

First-degree price discrimination

Under **first-degree price discrimination (FDPD)**, the firm charges each consumer the maximum price that he or she is prepared to pay for a good or service. It could occur in any market where there is some scope for bargaining over the price, eg gardening services. However, the firm would not usually be able to find out the maximum price that each consumer is willing to pay.

FDPD enables the firm to increase its profits to the entire area below the demand curve (D) and above the marginal cost (MC) curve (see diagram below, which assumes that MC is constant and fixed costs are zero, so that average cost (AC) = MC).

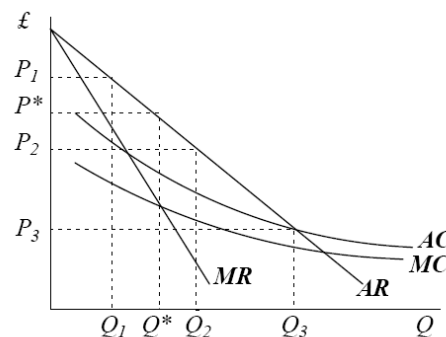
The rectangle $(P_1 - AC_1) \times Q_1$ represents supernormal profits for a firm that does not price discriminate. The two triangles represent the additional profits from price discrimination. FDPD will also lead to increased output, as the firm's demand curve in the absence of price discrimination effectively becomes its marginal revenue (MR) curve in the presence of FDPD, so it will produce where price = MC (as opposed to at $MR = MC$), at the socially efficient output level.

**Second-degree price discrimination**

Under **second-degree price discrimination (SDPD)**, the firm charges a different price to consumers depending on the number of units purchased. Typically, the average price paid falls with the number of units purchased.

Buy-one-get-one-free and “3 for the price of 2” offers are examples of second-degree price discrimination.

SDPD may encourage consumers to purchase more units than they would otherwise and so is useful when the firm's average costs decrease with output.



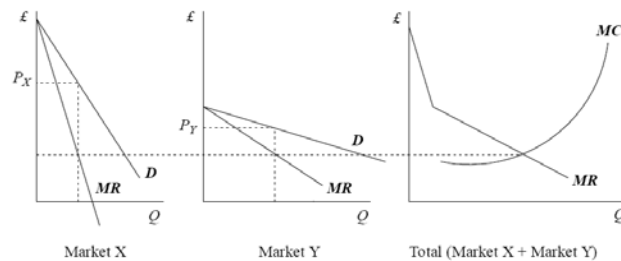
In the absence of price discrimination, the firm would charge P^* and output would be Q^* . With SDPD, the firm might choose to offer goods at P_1 , P^* , P_2 and P_3 (though other choices are possible) and so benefit from economies of scale and more revenue.

Third-degree price discrimination

Under **third-degree price discrimination (TDPD)**, a firm divides consumers into different groups and charges a different price to each group of consumers, though all the consumers within a particular group pay the same price.

Suppose a firm is selling an identical product in two separate markets, Market X and Market Y. Total profit is maximised when $MR = MC$ in the "Total" market. MR must be the same in both Market X and Market Y and the price in each market is then set from its respective demand curve. The price is higher in Market X where the demand curve is less elastic.

The price charged to each group will reflect the relevant elasticity of demand. A higher price will be charged where demand is more inelastic.



Examples of the three types of price discrimination:

First degree: Lawyers charging different fee from clients; Interior designer charger different fee from customers

Second degree: Buy-two-get-one-free; discount on bulk buying

Third degree: Different park entry fee for children and adults; different electricity rates fr commercial and domestic purposes

[6]

Solution 40 :

Barriers to entry prevent new firms from entering an industry. Barriers may occur naturally, or may be erected by the monopoly.

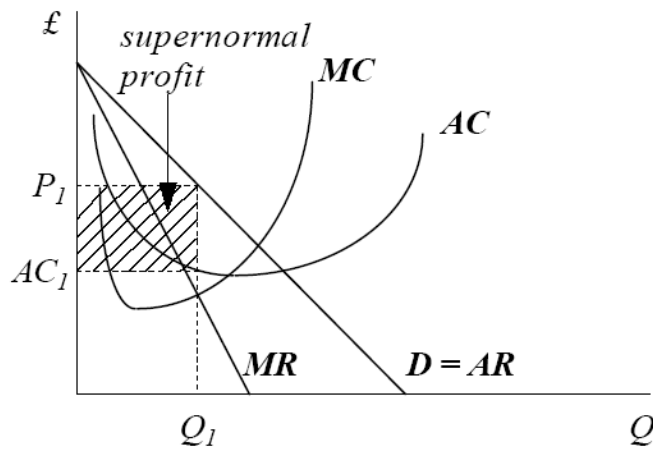
Natural (or innocent) barriers to entry include:

- *economies of scale*, especially where the minimum efficient scale is large (possibly so large that it satisfies the whole market)
- *economies of scope*, where the firm produces a range of products and can spread overheads across the range
- *lower costs for an established firm*, eg:
 - specialised production techniques
 - specialised marketing skills
 - relationships with cheapest / best suppliers
 - cheaper financing.

Strategic (or deliberate) barriers to entry include:

- *product differentiation and brand loyalty*
- *ownership or control over key inputs* (eg ownership of a key supplier) or *outlets*
- *threat of merger / takeover*
- *aggressive tactics*, eg selling loss leaders.

Legal protection may be either a natural barrier, eg requiring a licence to operate, or strategic, eg by patenting processes / products.



This diagram may apply in the short run or the long run. The only difference in short-run and long-run equilibrium for a monopolist is that in the long run, marginal revenue will be equal to long-run marginal costs.

Yes, it is possible for a monopolist to incur losses in the short run if its average cost becomes higher than the price consumers are ready to pay.

If average variable cost becomes higher than the price consumers are ready to pay, the monopolist should shut down production.

[4]

[Total Marks – 100]
